A browser-based seismic observatory for regional earthquake catalogue comparison in North America

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Abstract

Earthquake catalogues differ in coverage, measurements and revision histories, complicating regional comparison. Our browser-based observatory connects documented inputs, reproducible selections and reusable descriptive results. Four frozen USGS, NRCan and SSN packages supply 287,936 source records. Explicit adapters retain magnitude types, missing values and provenance without reconciling physical events. Verified static delivery supports interactive maps, statistics, rankings and exports without Docker or a database server. A 659-record USGS selection has mean magnitude 5.311; matched Canadian and Mexican land selections contain 15 and 53 records. Separately, a 36-event ComCat pilot finds magnitude changes in 13 of 31 comparable retained endpoint pairs and four crossings of 5.5. Independent SQL agrees with 4,720 conditional projections. Evidence contracts distinguish captured observations, archive-dependent calculations and unsupported historical reconstructions. Results establish neither completeness nor validated seismic risk.

Keywords: earthquake catalogues; geovisualization; provenance; retained revisions; reproducibility; evidence contracts

1. Introduction

Regional comparison requires duplicate handling and compatible magnitudes (Weatherill et al., 2016); hazard analysis additionally needs completeness assessment and declustering (Sawires et al., 2019). A multi-catalogue map alone supplies neither.

We address the conversion of heterogeneous regional records into inspectable selections and reusable descriptive information. The observatory connects browser computation, documented seismic inputs and traceable results for further research, using frozen snapshots rather than live monitoring.

ComCat products evolve (U.S. Geological Survey, n.d.a-c; Thompson et al., 2020). Existing history tools include Libcomcat, Timeline and INGV (U.S. Geological Survey, n.d.d, e; Pintore, 2026). Our contribution combines regional exploration with executable evidence contracts, informed by resolution-aware queries (Sismanis et al., 2009; Gardezi and Bertossi, 2011). Inputs extend Villa Vargas et al. (2026); Figure 1 provides the working architecture.

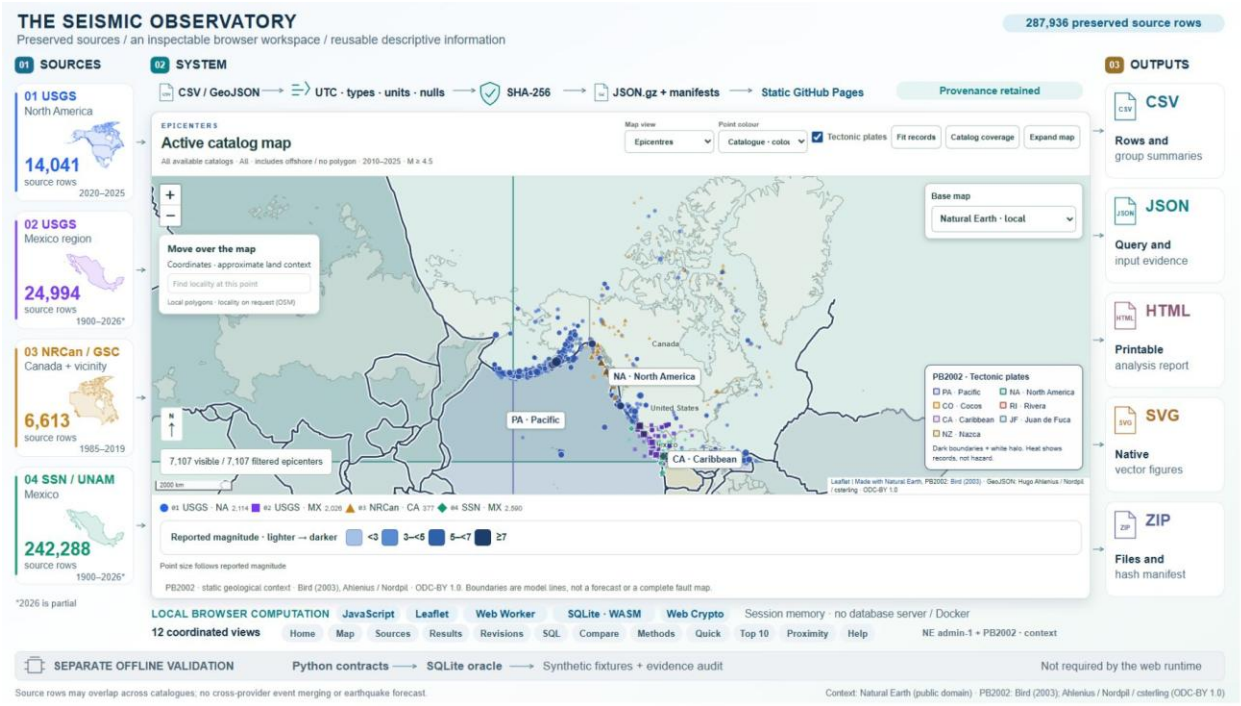


Figure 1. Implemented architecture connects four source packages, explicit adapters and verified static delivery to browser maps, analysis and reusable exports. The shared interactive map runs inside the design and shows 7,107 selected rows (2010-2025, $M \geq 4.5$, all catalogues); inputs total 287,936. Optional SQLite/WebAssembly runs in a Worker; country/subdivision indexes and PB2002 supply separate context. The offline Python/SQLite audit is distinct from the browser runtime. Interactive view: Open implemented design.

2. Computational system and analytical methods

2.1. Static architecture and reproducible delivery

Python adapters produce JSON and SHA-256 manifests from preserved CSVs. GitHub Pages serves HTML/CSS/JavaScript, Leaflet 1.9.4 and local dependencies; verified records occupy session memory without Docker or a database server. Figure 1 embeds the shared working map, not a separate image or duplicate application.

A loading gate enables interaction after verification and rendering. Home selects all 287,936 rows (Figure 2); English, Spanish and appearance controls are available.

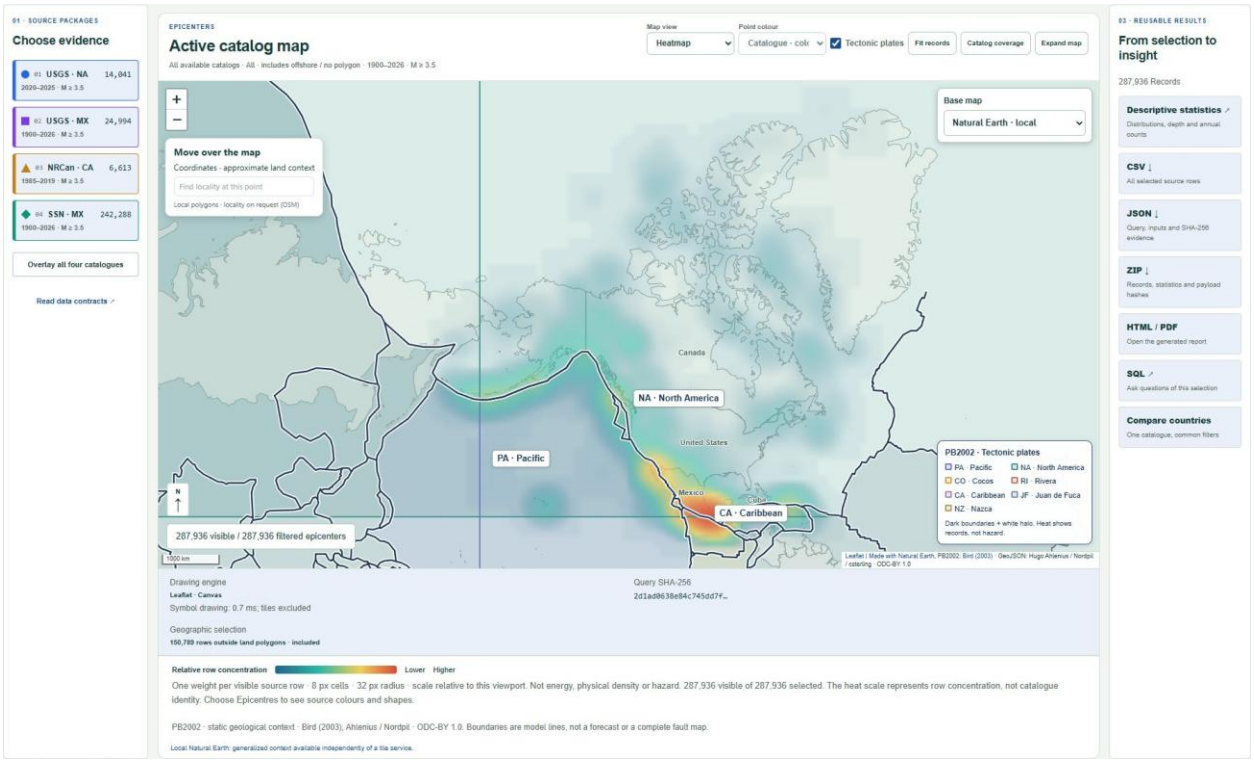


Figure 2. Home initially selects all 287,936 rows across four catalogues. Point and optional heat views retain the same query and exports. Heat displays relative screen-space row concentration, without magnitude weighting or hazard estimation. Labelled PB2002 boundaries render above heat; spatial coincidence establishes no mechanism. Interactive view: Open this view.

Compressed and original hashes/lengths are verified before parsing. Progress reports actual transfers; retries are bounded. Unsupported gzip or HTTP 404/415 allows verified JSON fallback, whereas integrity failures stop loading. Session caching avoids repeated downloads.

2.2. Normalization and geographic selection

Adapters align IDs, labelled UTC, coordinates, reported magnitudes and documented depths (Figure 3), retaining raw values, case, empty strings and unknowns; absent common fields become null. No Mw conversion occurs. Undocumented Canadian depths are excluded from kilometre statistics. ComCat IDs survive; SSN/NRCan use content-hash row keys with occurrence suffixes. Hue/shape identifies catalogue; the visible lightness legend encodes M <3, 3-<5, 5-<7 and ≥ 7 without scale equivalence.

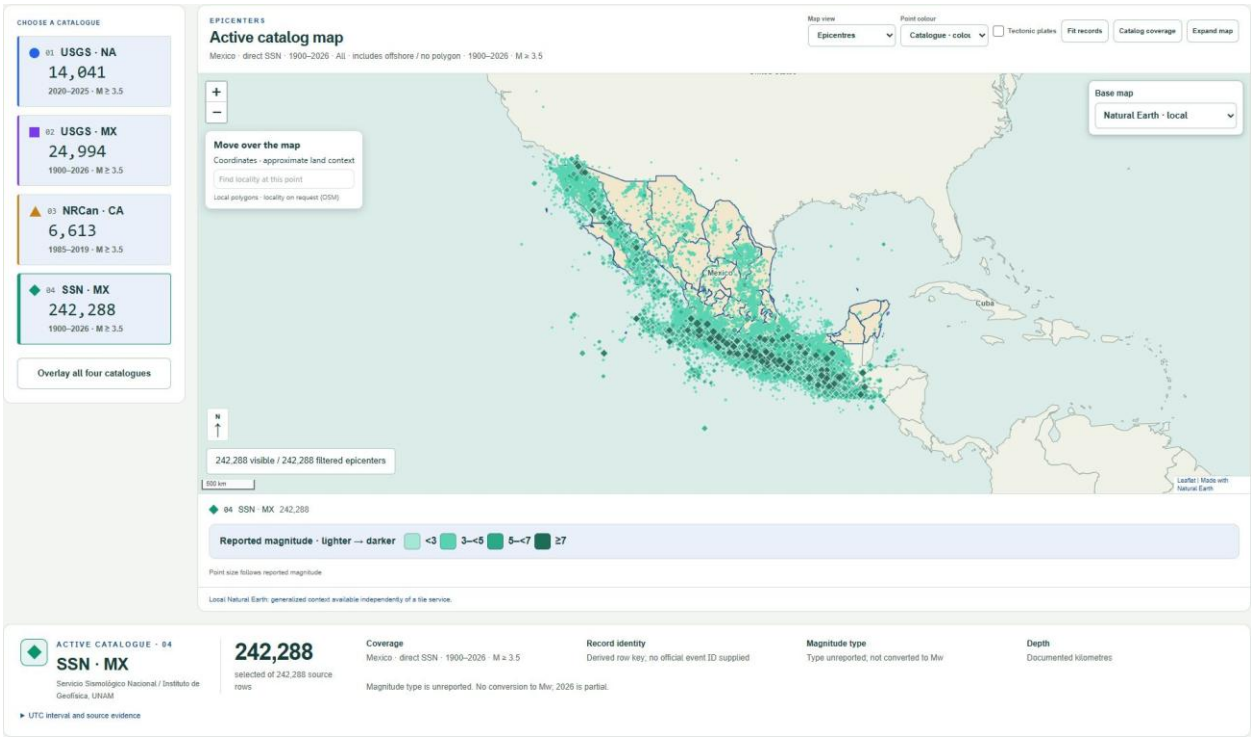


Figure 3. Sources Atlas displays the SSN package (242,288 rows), a regional map and a four-source rail. The complementary Time and Coverage view is accessible through its button. Hue/shape identifies source; darker tones indicate larger reported magnitudes within that hue. Four packages from three distributors are not independent observing agencies. Interactive views: Atlas · Time and coverage.

Composition preserves registry order and qualified keys. It flags 2,163 shared ComCat IDs without merging rows or matching agencies by proximity; 287,936 rows are not asserted independent earthquakes.

Pinned Natural Earth indexes bind geometry, event hashes and row order: 47 land classes and 743 subdivisions (Natural Earth, n.d.a, b), including Mexico's 32 named units and minor-island unit. Confirmed country clicks enable subdivision selection; Mexican boundaries also display without land filtering (Figure 4). Offshore, boundary and unavailable classes remain distinct. Generalized containment establishes neither jurisdiction nor nationality.

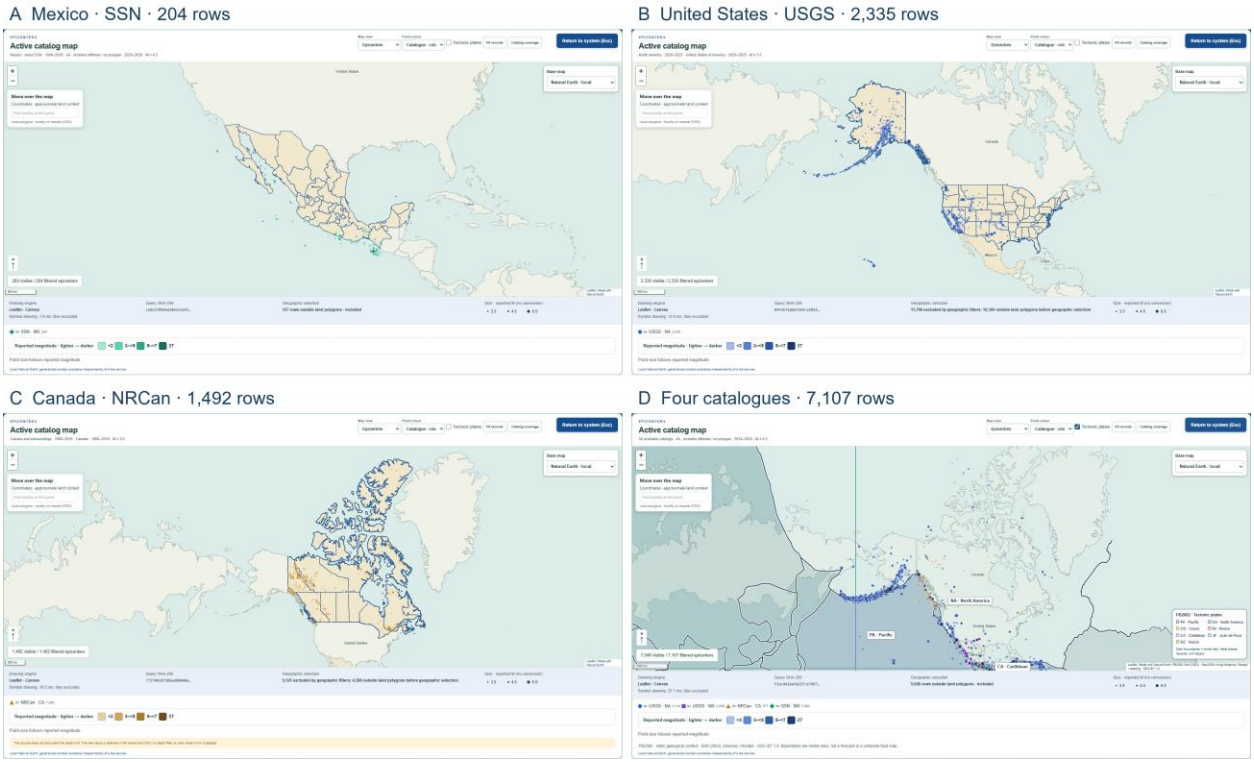


Figure 4. Four distinct map selections. (A) SSN, partial 2026, M ≥ 4.5, all classes: 204 rows, with Mexican subdivision context but no land filter. (B) USGS North America, 2020-2025, M ≥ 3.5, USA land: 2,335. (C) NRCan, 1985-2019, M ≥ 3.5, Canada land: 1,492. (D) All catalogues, 2010-2025, M ≥ 4.5: 7,107. Counts are source rows; map extent does not restrict exports. Interactive views: A · B · C · D.

Figure 5 contrasts subdivision selections under a common requested interval; archive gaps remain explicit.

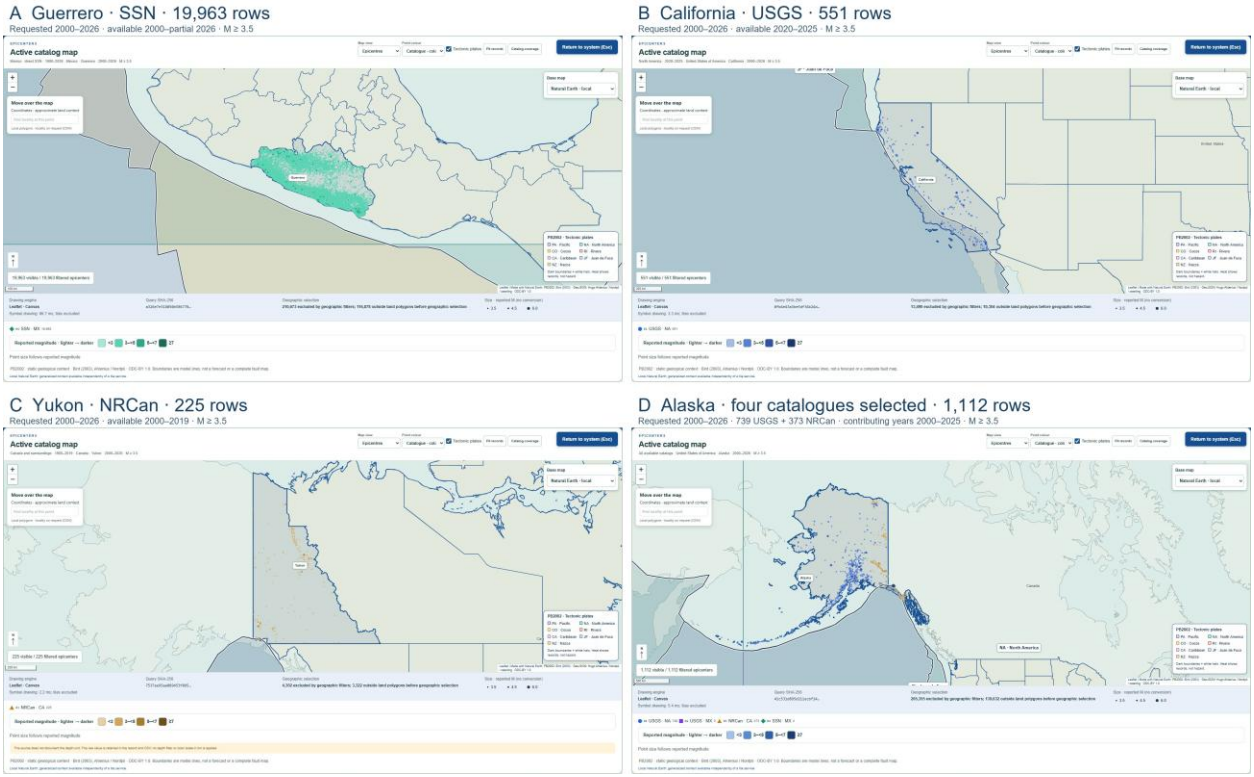


Figure 5. Subdivision queries, requested UTC 2000-2026, reported M ≥ 3.5 and land containment: (A) Guerrero, SSN, 19,963 rows (2000-partial 2026); (B) California, USGS North America, 551 (2020-2025); (C) Yukon, NRCan, 225 (2000-2019); (D) Alaska, four catalogues selected, 1,112 (739 USGS + 373 NRCan; 2000-2025). Archive limits remain visible; these are not complete 27-year national histories. Interactive views: A · B · C · D.

Filters cover UTC, magnitude/type, agency, documented depth, missing error and ID/place. Coordinated views follow Wu et al. (2020). Local Natural Earth and four remote backgrounds retain attribution. Viewport/longitude display changes preserve selection and exported coordinates. A query digest binds inputs, predicates, geography and ordered keys.

Heat mode assigns equal row weights to visible 8-pixel CSS cells, smooths with a 32-pixel Gaussian radius (sigma 12) and maps log1p values relative to the viewport maximum, clipping edge kernels. It represents neither physical density nor hazard. PB2002 boundaries and labelled plates render above heat, with a separate legend. Local polygon context and optional cached OSM lookup are approximate. USGS M ≥ 6 links open official ShakeMap/DYFI products; magnitude supplies no inferred felt radius.

2.3. Descriptive tools and historical queries

Statistics remain separate by catalogue and exact reported magnitude type; unknown types form their own numerical group. Means, medians, ranges and sample SD retain valid counts. Depth requires documented kilometres; Pearson correlation requires a known type, three finite pairs and nonconstant variables. Unavailable results remain null; annual zeros mean no selected rows.

Within USGS North America, country comparison shares UTC/type/threshold and 0.25-M histogram bins, without completeness or area/population normalization.

Top 10 queries all four catalogues, refined by country/state and optional type/period. Exact shared ComCat IDs yield one entry with both provenances; before filtering, the later valid update wins, with invalid/tied times preferring North America then Mexico, never larger magnitude. SSN/NRCan remain unmatched. Bounded selection orders descending M, ascending UTC and key, omitting additional ties. Offset labels preserve original epicentres. Rankings are heterogeneous reported values, not homogeneous Mw or the strongest-ever population.

Historical proximity enumerates distinct-key pairs within one catalogue/type/UTC interval, default $M \geq 7$, using Haversine R=6,371.0088 km and 25/50/100/200-km radii. Optional 100-km offshore context uses minimum spherical point-to-minor-arc land distance, without annexing assigned neighbours. A 10^{-9}-km numerical tolerance precedes rounding. Direct pairs imply neither transitive clusters nor repeated ruptures, recurrence intervals or risk. PB2002 supplies static context, not mechanisms or active faults (Bird, 2003; tectonicplates contributors, n.d.).

Optional sql.js 1.13.0 runs isolated SQLite/WebAssembly queries (sql.js contributors, n.d.): qualified keys/nulls, one SELECT/WITH, cancellation, timeout and 5,000-row limit.

Simple view shares the full engine; Help supplies coverage/contact/licensing. CSV/JSON/HTML/SVG retain scope, ZIP adds hashes, and printable HTML supports PDF. Query schema seismic-local-query/5 preserves geographic evidence and older URLs.

2.4. Contracts for retained revisions

The offline audit identifies lineages by (source, type, code) and versions by updateTime. Exact repeats deduplicate; conflicting content fails. Distributor, origin and magnitude authors remain distinct. Primary endpoints require source us, type origin and code matching the captured event ID; internal-origin/phase-data are excluded.

Table 1 defines the contracts. Projection selects each lineage's latest retained producer time at/before a cutoff, handles DELETE without reviving predecessors, then filters. Filtering first can retain obsolete qualifying versions. Nulls fail predicates; literal types are not converted.

Table 1. Contract	Interpretation of the output	Principal boundary
`snapshot_observed`	Query result on supplied current summary bytes at capture	Not a historical replay or a complete physical population
`retained_projection_conditional`	Latest retained version per lineage at or before a producer-time cutoff	Neither public availability nor historical preference is established
`historical_catalog_not_identifiable`	Stronger historical catalog claim is declined	Missing historical evidence is not manufactured

Unknown membership, release times, preferences and associations prevent complete historical public-state reconstruction. Retained-only evidence and producer/publication clocks remain explicit; declining a stronger claim does not invent missing versions.

Endpoints are earliest/latest retained non-deletions, not first release or truth. Subtraction requires identical literal magnitude types; inclusive thresholds are 5.5-8.0 in 0.5 steps. Haversine measures separation, not accuracy gain. Adjacent versions compare 27 declared parameter/agency/uncertainty fields.

2.5. Independent evaluation

Independent SQLite normalizes original products and ranks before filtering, using Decimal magnitudes and integer UTC times. Ten specifications test thresholds, mww/mb/mB, bounds, earthquake type and half-open 2017/2022 intervals immediately before/at updates. Hashes and lengths bind inputs.

Three paired histories share current evidence but differ in past release, preference or withdrawal. Eight further histories test missing revisions, deletion and clock/filter substitutions; regional/magnitude variants yield 24 cases with independent replay. They demonstrate ambiguity, not prevalence. Comparator documentation establishes context, without performance/usability benchmarks; Timeline access failed.

3. Seismic inputs and study design

Four catalogue packages from three distributors demonstrate exploration separately from revision validation (Table 2). Their coverage is not a sequence of historical public states.

Table 2. Catalogue and distributor	Origin-time coverage	Records	Interpretation limits
Mexico-region diagnostic / USGS ComCat	1900-2026 (partial)	24,994	Earthquake records; regional rectangle; two quarry blasts excluded; not an independent SSN catalogue
North American regional windows / USGS ComCat	2020-2025	14,041	Earthquake records from twelve pinned CSVs; regional windows, not national assignments or a revision archive
Canada and surroundings / NRCan/GSC	1985-2019	6,613	Reported M ≥ 3.5; five repeats retained; original event IDs, event types and uncertainty/revision fields absent
Mexico and surroundings / direct SSN	1900-2026 (partial)	242,288	Reported M ≥ 3.5 selected from 402,082 source rows; magnitude types, original IDs and revision fields absent; coverage is not completeness

Mexican packages include partial 2026. USGS regional/Aleutian windows span latitude 14-84°, longitude -180 to -50°, plus Aleutians 50-73°/170-180°. Original positive longitudes survive in 770 rows.

Canada retains 6,613/101,365 rows at M ≥ 3.5, including five repeats (6 November 2026 download). IDs/types/depth units are unknown. Its Open Government Licence-Canada differs from NEDB terms (Natural Resources Canada, n.d.; Government of Canada, n.d.).

SSN covers 1900-2026-11-06, all magnitudes/depths/states (Servicio Sismológico Nacional, 2026a). Ten fields preserve UTC/local times, magnitude, coordinates, kilometre depth, locality and status. Of 402,082 rows, 384,433 have numeric magnitude and 242,288 satisfy M ≥ 3.5; 17,649 are uncalculable. IDs, types, uncertainty and revisions are absent; verificado/revisado are statuses.

The SSN generation label implies 19:57:07 UTC, before the latest origin (20:31:07); eight UTC/local pairs have atypical offsets. Both columns and tied annual maxima remain unchanged; the label is not an atomic capture. Credits/terms are retained (Servicio Sismológico Nacional, 2026b). The documented 1988 reporting change and M_d 4.3 completeness for 1988-1998 (Zúñiga et al., 2000) do not establish completeness here.

The 46-CSV upstream snapshot 8fe055ffe635875f7290906d45f70562a7896621 contains 24,996 IDs; two quarry blasts are excluded. UNM/us_unm selects 9,858 earthquakes with SSN/UNAM authorship distributed by USGS (European-Mediterranean Seismological Centre, n.d.).

The separate ComCat pilot selects current 2017/2022 earthquakes with preferred M ≥ 5.5, latitude 14-33° and longitude -118 to -86°. Retrieval spans 5 November 2026, 19:53:15.316-19:53:54.609 UTC. Count/detail responses (format=geojson, includesuperseded=true) retain bytes, URLs and hashes. Today's selection cannot identify all historical withdrawals/crossings.

4. Results from reproducible selections

4.1. Regional information and reuse

Source profiles expose coverage and verification. Figure 4 contrasts SSN, USA-land, Canada-land and combined selections; distributor and geographic containment remain separate.

The guided SSN Guerrero land selection (UTC 2020-2025, M ≥ 3.5) contains 7,930 rows: mean M 3.691, median 3.6 and mean documented depth 31.17 km (Figure 6). Cards, histogram and six annual bins share the selection; absent magnitude types do not establish scale homogeneity.

An independent 659-row USGS benchmark (2020-2025, M ≥ 4.5, mww, all classes) yields mean M 5.311168, median 5.2, range 4.5-8.2 and sample SD 0.523513. Depth mean/median are 32.832339/15.14 km; Pearson r=-0.003260 establishes neither independence nor causality. Python matches JavaScript and ordered IDs within 10^{-12}; exports retain criteria/hashes.

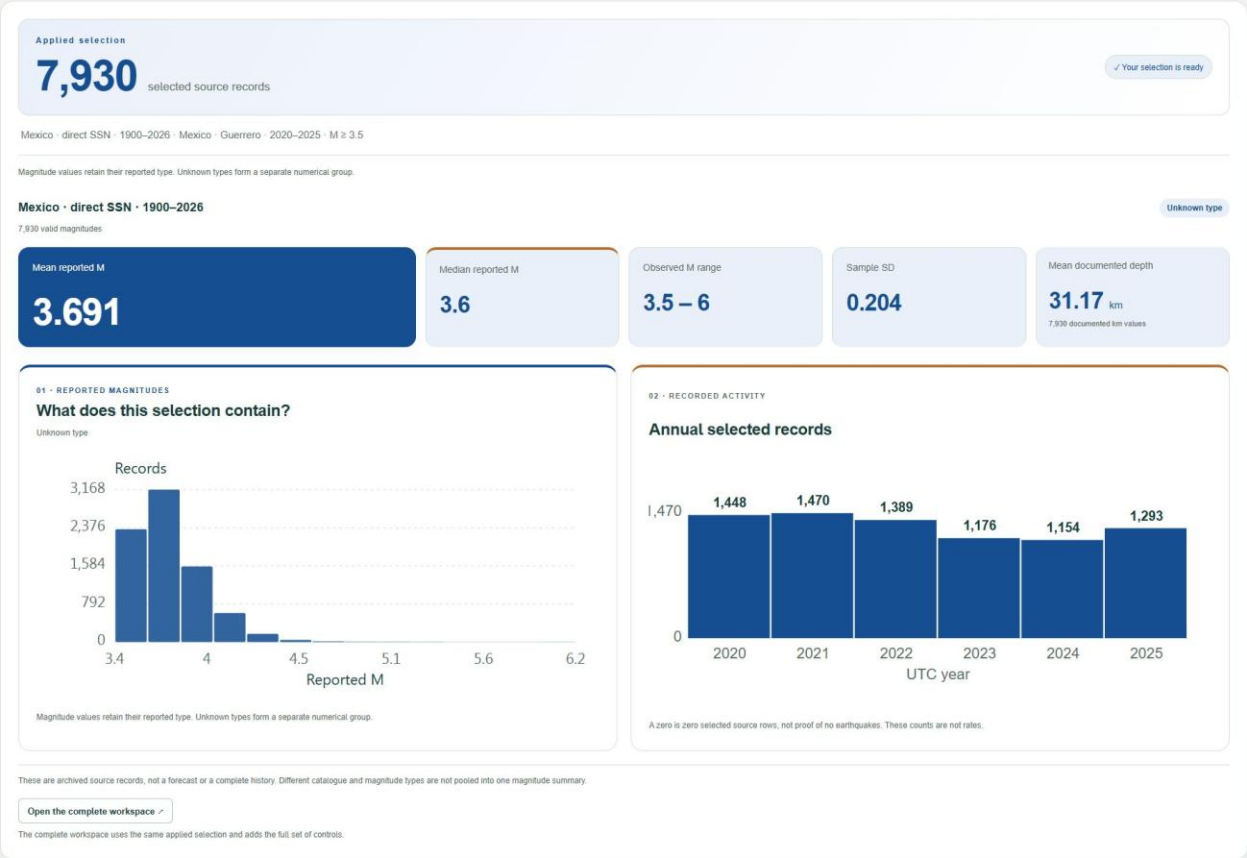


Figure 6. Guided descriptive results for 7,930 SSN Guerrero land records, UTC 2020-2025, $M \geq 3.5$. Summary cards, the 0.2-M histogram and six annual bins use the same selection. Reported magnitude types are absent; numerical grouping establishes neither scale homogeneity, completeness nor physical independence. No occurrence-rate or risk estimate is presented. Interactive view: Open these results.

Administrative selection changes the query; plate display does not. Figure 7 enlarges 203 Guerrero SSN land rows (2000-partial 2026, $M \geq 4.5$), without supplying mechanisms or location uncertainty.

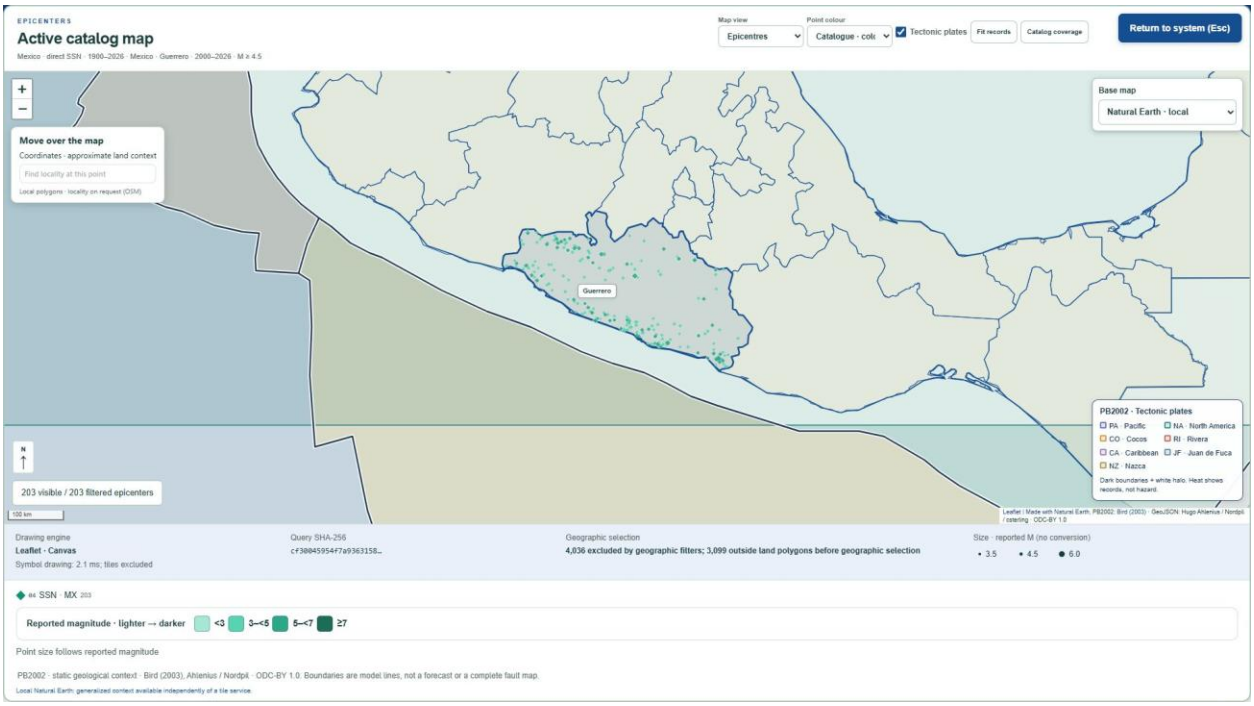


Figure 7. The enlarged Guerrero map shows 203 SSN land rows, UTC 2000-2026 (partial), M ≥ 4.5, administrative limits and static PB2002 context. Confirmed country/subdivision filters alter the query; plate display does not reclassify records or supply mechanisms. Natural Earth classes are generalized land containment, with unassigned/offshore rows retained separately. Interactive view: Open this zoomed map.

Matched USGS period/threshold/mww selections contain 15 Canadian and 53 Mexican land rows (Figure 8). Shared bins describe selected records; unequal areas, excluded offshore rows and unassessed completeness preclude relative-hazard or occurrence-rate comparisons.

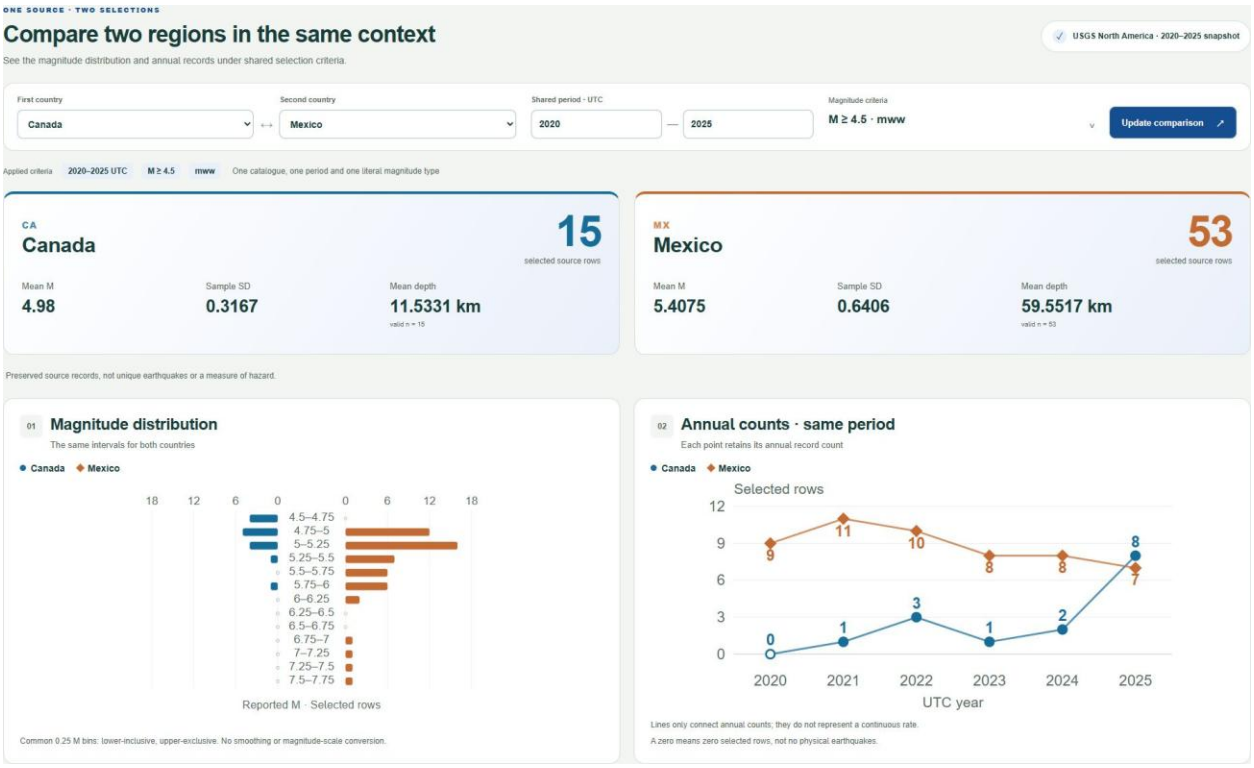


Figure 8. Canadian and Mexican land selections within USGS North America, UTC 2020-2025, M ≥ 4.5, literal mww: 15 and 53 rows. Common 0.25-M bins and annual counts preserve selection/evidence; they are not completeness-adjusted rates or relative hazard. Interactive view: Open this comparison.

4.2. Guided historical consultation

Simple view retains the complete workspace and shared regional/export engine (Figure 9). Help explains coverage and terms.

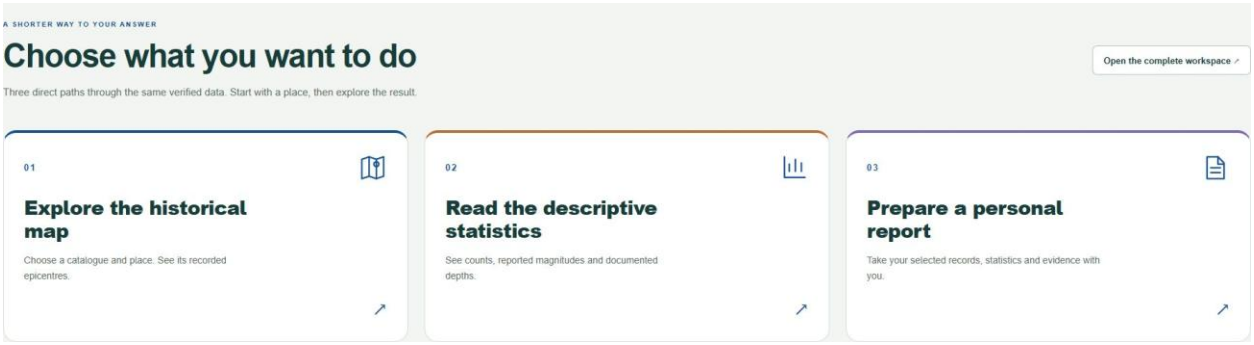


Figure 9. Three implemented Simple-view routes open historical maps, descriptive statistics and personalized reports through the complete workspace's filter/export engine. The full interface remains accessible. The capture demonstrates controls rather than measured usability gains. Interactive view: Open Simple view.

Top 10 preserves four-source measurements (Figure 10 A). In SSN-only mode, Guerrero contains 21,167 M ≥ 3.5 land rows (1900-partial 2026, unknown type); its top entry is M 7.8 on 28 July 1957. Fourteen rows satisfy M ≥ 7, yielding ten ≤50-km pairs involving twelve (Figure 10 B), without establishing independent repeats or hazard. An independent Python geometric oracle checks ranking/distance cases.



Figure 10. Complementary historical map extracts. (A) Default Top 10 ranks reported magnitude numbers across four catalogues, retaining source/type and both provenances for shared exact ComCat IDs; these are not homogeneous Mw measurements or reconciled physical events. Rank labels may be offset, with leader lines ending at the original epicentres. (B) SSN Guerrero land, UTC 1900-2026 (partial 2026, unknown magnitude type): fourteen rows satisfy M ≥ 7; ten pairs at ≤50 km involve twelve. Offshore context is disabled. Links/circles are selection geometry, not rupture recurrence or risk. Tables and CSV/JSON preserve criteria and hashes. Interactive views: A · Top10 · B · proximity.

4.3. Retained revisions and source roles

The 36-event pilot (22 in 2017;14 in 2022) retains 236 UPDATE origin versions, including 205 primary US versions (Table 3). Retention does not imply simultaneous activity; ISC-GEM illustrates distribution/authorship differences.

Table 3. Observed quantity	Value	Counting unit
Current-selected event identifiers	36	Captured events
Public origin-product versions	236	Retained versions
Versions in primary US origin lineages	205	Retained versions
Distributor source us	226	Public origin versions
Distributor source at / pt	6 / 4	Public origin versions
Origin author us / iscgem / missing	205 / 21 / 10	Public origin versions
Separate internal-origin / phase-data versions	15 / 203	Related products, excluded from primary analysis

Endpoint displacement has median 12.339 km and maximum 79.684 km; depth differences span -38.48-12.39 km. For us2000ahv0, retained endpoints change 8.0→8.2 mww, 35.0→47.39 km depth and 9.938 km epicentrally, without establishing accuracy.

Thirteen of 31 same-type pairs change magnitude; five mixed-type pairs are excluded. Four cross 5.5 (Table 4), none higher tested thresholds. Among 30 mww pairs, earliest/latest endpoints select 26/30 at M ≥ 5.5. Current-cohort admission prevents interpreting upward crossings as an unbiased migration rate.

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ailq	5.4	5.5
us2000ap3b	5.4	5.6
us2000aqji	5.4	5.5
us1000arzf	5.4	5.5

Of 169 adjacent primary pairs, 117 change selected fields. Ten summary timestamps postdate every retained origin. For us2000ahv0, the primary branch ends in December 2017, another updates in May 2022, and the October-2026 summary concerns DYFI. These clocks do not form one origin-revision sequence.

4.4. Validation and observed coverage

At 236 update boundaries, 472 cutoffs produce 4,720 queries agreeing with independent SQL; all 24 synthetic conditional projections agree. Against synthetic public truth, backcasting, filtering first and index-time substitution fail in 10/11/10 overlapping cases: constructed failures, not operational rates. Stronger claims are declined; empirical products contain no DELETE cases.

SSN occupies 110/126 closed years (1900-2025), leaving 16 empty bins. Eight tied years yield 118 maximum-attaining rows. Partial 2026 contains 34,548 raw rows, 14,875 at $M \geq 3.5$ and 35 at $M \geq 5$. Coverage establishes neither completeness nor rates/return periods.

5. Discussion

Reuse preserves heterogeneous scales and multiplicity. Land filters exclude offshore records; nearby context retains unassigned status. Spatial pairs, correlations and plates establish neither rupture recurrence nor risk.

Evidence contracts bound calculations when past public states are unknown. Prior history software remains relevant; no superiority is claimed. Synthetic counterexamples demonstrate ambiguity; SQL independently validates the evidence.

Static files aid migration but guarantee neither twenty-year hosting nor browser compatibility. Memory/service limits persist. Responsive checks are not usability evidence; future work needs independent users and historical captures. Probabilistic models require justified completeness, scale, sampling and stationarity.

6. Conclusions

Browser computation connects documented inputs and traceable descriptive results without a database server. Validation establishes reproducible calculations, not source accuracy, completeness or historical public states. No earthquake forecast or validated risk estimate is produced.

Code and data availability

The evaluated interface is release 2026.11.11.2. Distribution uses the study repository (https://github.com/gabrielhuav/Seismic_Data_Geosciences) and its public explorer (https://gabrielhuav.github.io/Seismic_Data_Geosciences/). Software, manifests and frozen evidence accompany validation. Source hashes and capture receipts identify evaluated inputs and images; no archival DOI is asserted. All eighteen article targets restore the English selections shown in the figures.

The archive preserves 36 detail and four count/query responses; restore verified bytes rather than substitute live endpoints. SSN's full CSV SHA-256 is 0601c65544bc95ef865d3440e5e6d84249acdebd8edfb379669597096543f993. Its DOI identifies a mutable service, not this download. Credit belongs to SSN, Instituto de Geofísica, UNAM, without endorsement. Source-specific terms remain separate from MIT software, sql.js/SQLite notices, public-domain Natural Earth and ODC-BY PB2002 attribution.

Run python -m unittest discover -s tests and python -m scripts.validate_evidence; documented JavaScript/oracle checks validate browser calculations. Manifests distinguish observed inputs, geographic derivatives and synthetic controls.

CRediT authorship contribution statement

José Manuel Villa Vargas: Conceptualization; Formal analysis; Supervision; Writing - original draft; Writing - review & editing.

Gabriel Hurtado Avilés: Data curation; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI assistance

OpenAI Codex (OpenAI; GPT-6-based agent sessions, 6-11 November 2026) assisted literature discovery, code development, reproducible figure scripts, automated testing, manuscript drafting and document formatting. The explanatory diagram and data plots are rendered from documented inputs and maintained code. The interface panels derive from genuine integrated-browser captures of local previews, with capture-specific dates and hashes retained. Cropping and panel composition preserve the captured content; no generative image creation or retouching of observations is used.

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